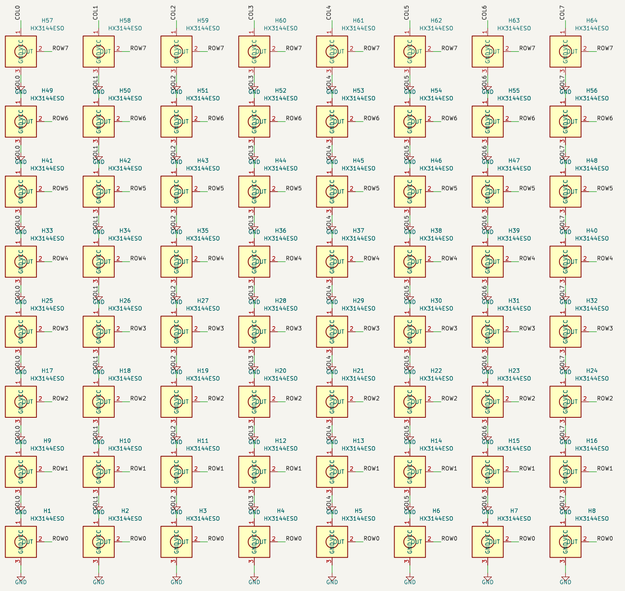
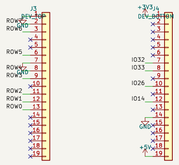
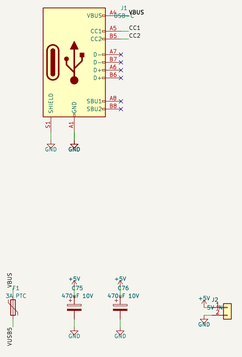
HALL SENSOR MATRIX (columns = switched +5V via PNP, rows = open-collector OUT wired-OR, 10k pull-up to 3V3)



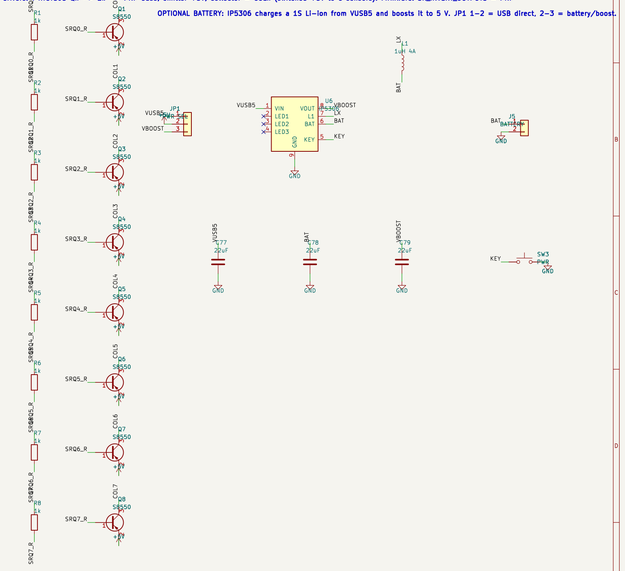
CONTROLLER (ESP32-WROOM-32E, CH340C USB-UART, AMS1117-3.3, 74HCT595 column driver, 74AHCT1G125 LED level shifter)



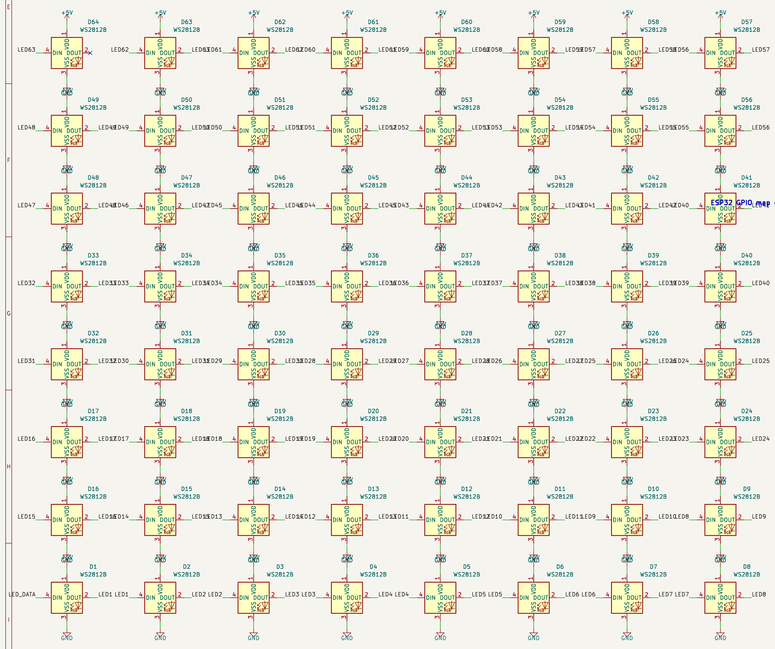


Column drivers: 74HCT595 Qn -> 1k -> PNP base; emitter +5V; collector = COLn (switched +5V to 8 sensors). Firmware: SR_INVERT_OUTPUTS = PNP

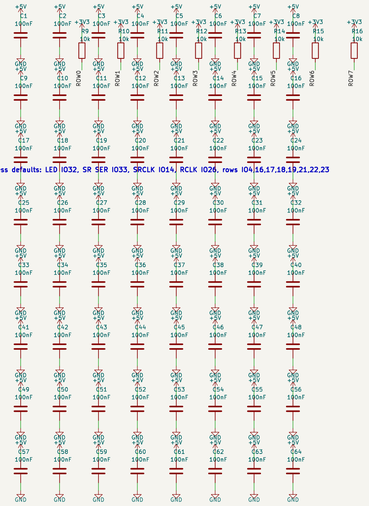
OPTIONAL BATTERY: IP5306 charges a 1S Li-Ion from VUSB5 and boosts it to 5 V. JP1 1–2 = USB direct, 2–3 = battery/boost.



WS2812B LED CHAIN (serpentine a1->h1, h2->a2, ... ; each LED has a 100nF cap)



ESP32 GPIO map = OpenChess defaults: LED IO32, SR SER IO33, SRCLK IO14, RCLK IO26, rows IO4,16,17,18,19,21,22,23



POWER FLAGS



pitch 40.0 mm; sensor HX3144ESO pinout {'VCC': '1', 'OUT': '2', 'GND': '3'}
Generated by tools/gen_sch.py — firmware-compatible with joojooo0/OpenChess

Sheet: /	
File: openchess-board-nodemcu.kicad_sch	
Title: OpenChess PCB redesign — 8x8 Hall/WS2812B board (nodemcu variant)	
Size: A1	Date: 2026-10-21
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Rev: 1.0
	Id: 1/1